

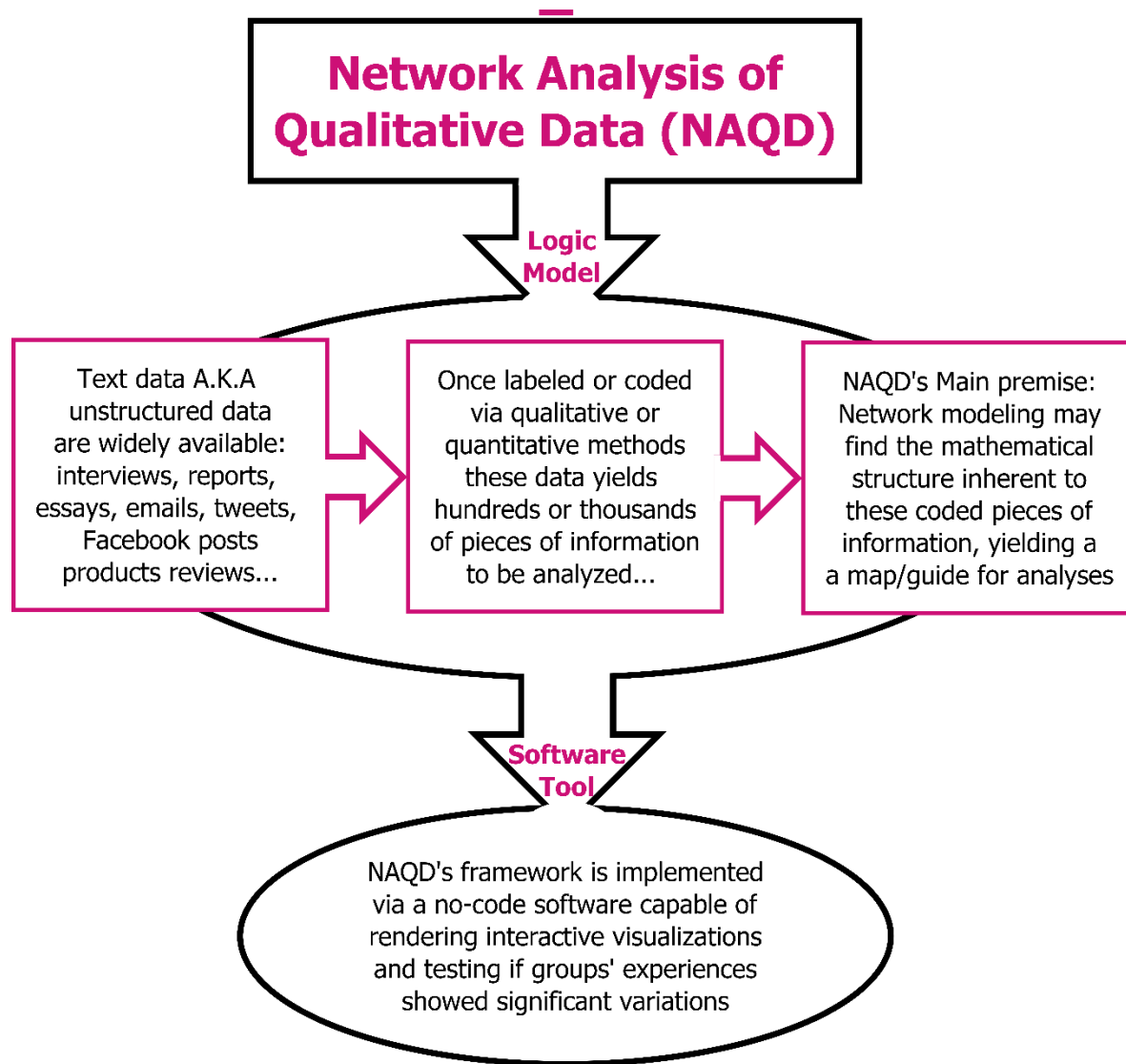


Figure 1. Conceptual Framework

This is the edgelist
representation of
Table 1

	A	B	C
1	actor	code	role
2	ID1	Professional Development	Student
3	ID1	Hands-on Experiences	Student
4	ID1	Promotion Structure	Student
5	ID2	High Program Impac	Faculty
6	ID2	High Program Impac	Faculty
7	ID3	Hands-on Experiences	Administrator
8	ID3	Professional Development	Administrator

This is the network
depiction of that
edgelist

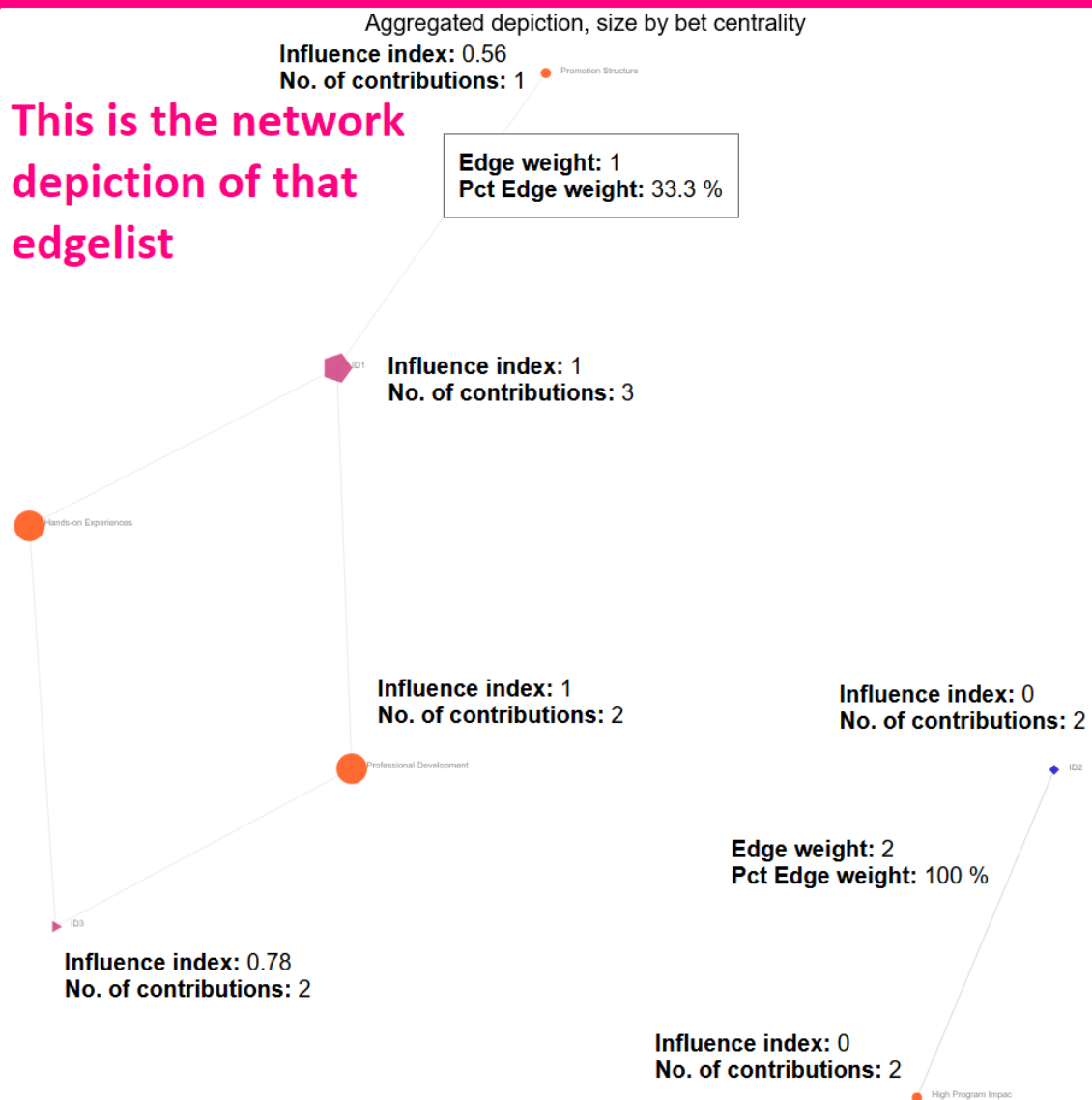


Figure 2. From Edgelists to Sociograms

	Observed Matrix 1			Observed Matrix 2		
	U1	U2	U3	U1	U2	U3
U1	2	1	5	4	2	5
U2	1	4	0	0	5	0
U3	1	0	6	2	0	7
	Observed Matrix 1			Shuffled Matrix 2		
U1	2	1	5	?	?	?
U2	1	4	0	?	?	?
U3	1	0	6	?	?	?

Figure 3. Matrix Manipulation Behind Quadratic Assignment Procedures

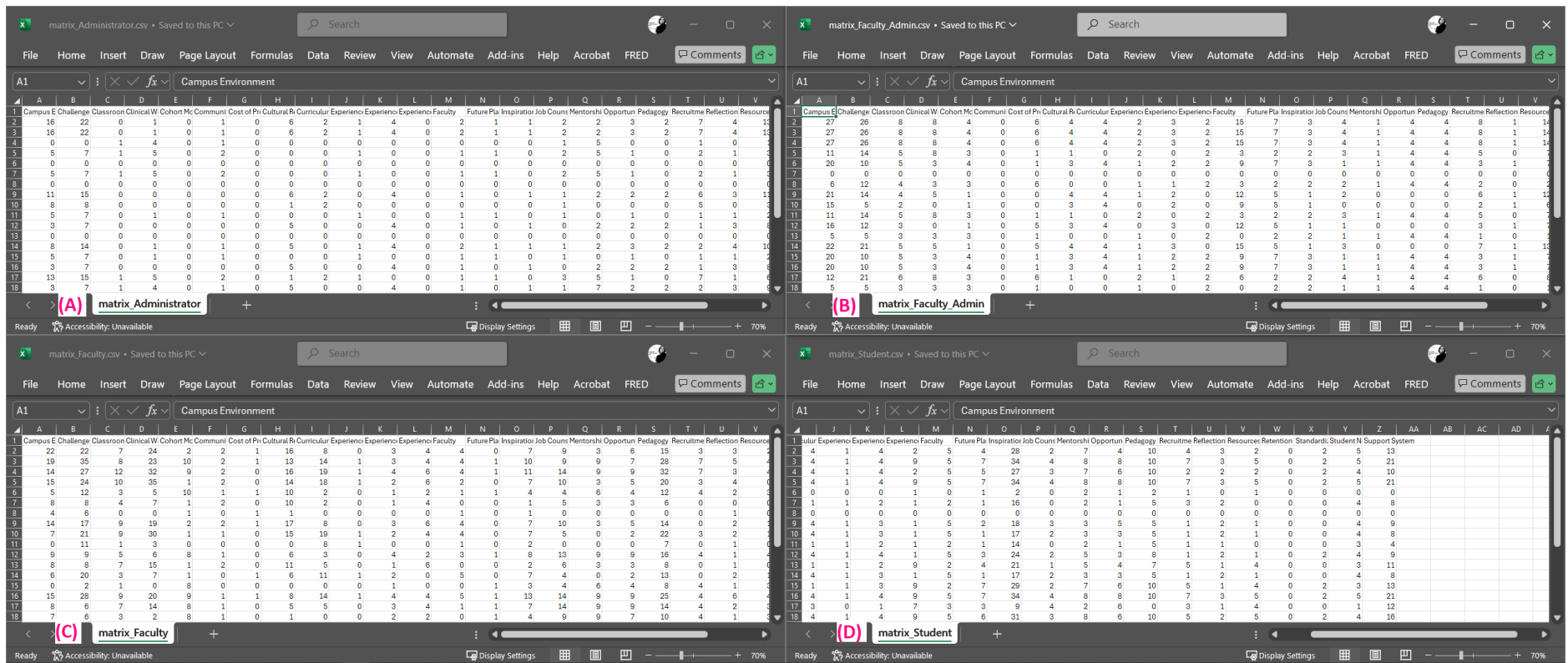


Figure 4. Projected One-Mode Matrices

Aggregated depiction, size by bet centrality

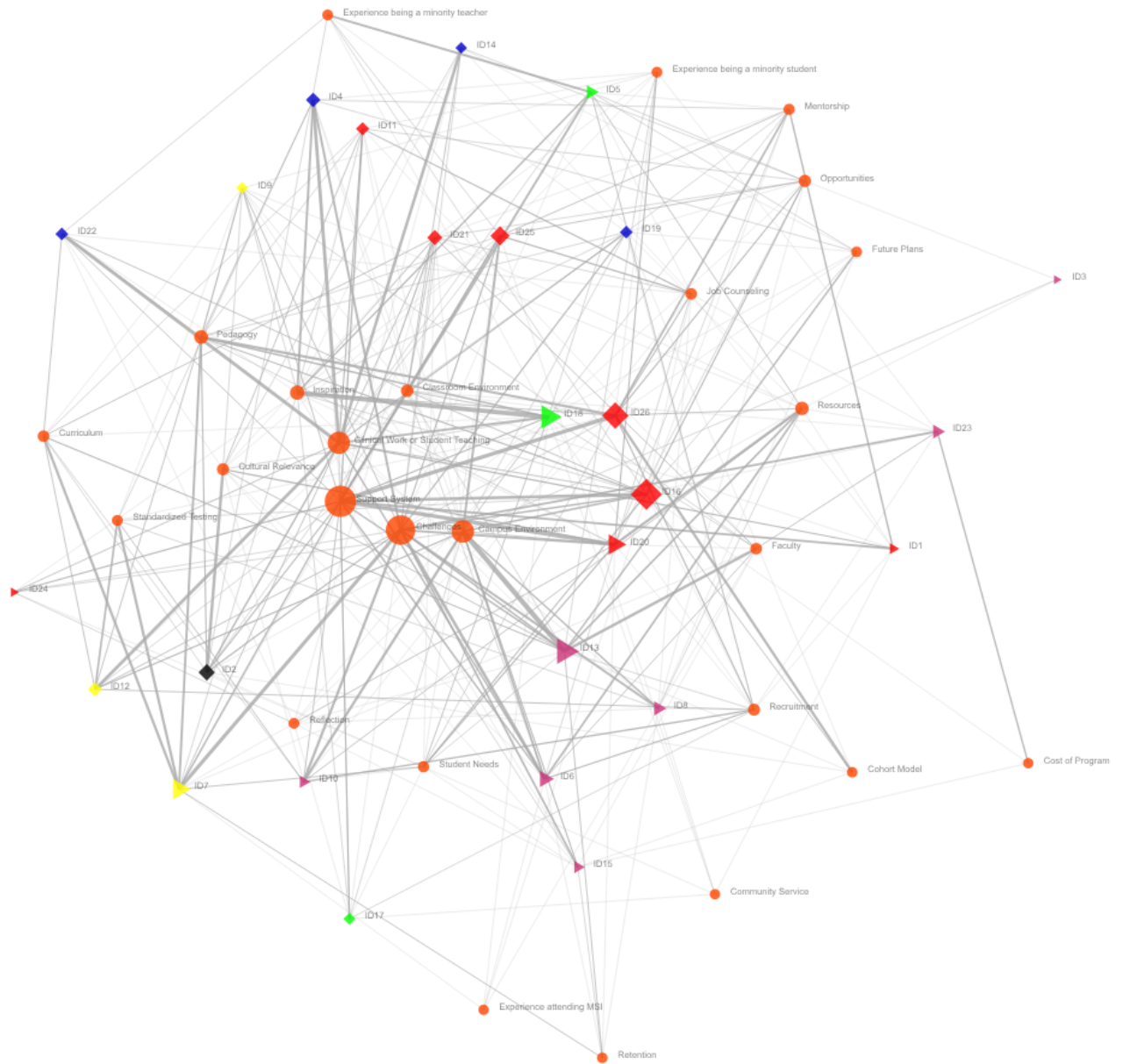


Figure 5. Sociogram of Aggregate Connections (see interactive version here <https://cutt.ly/8rikqLil>)

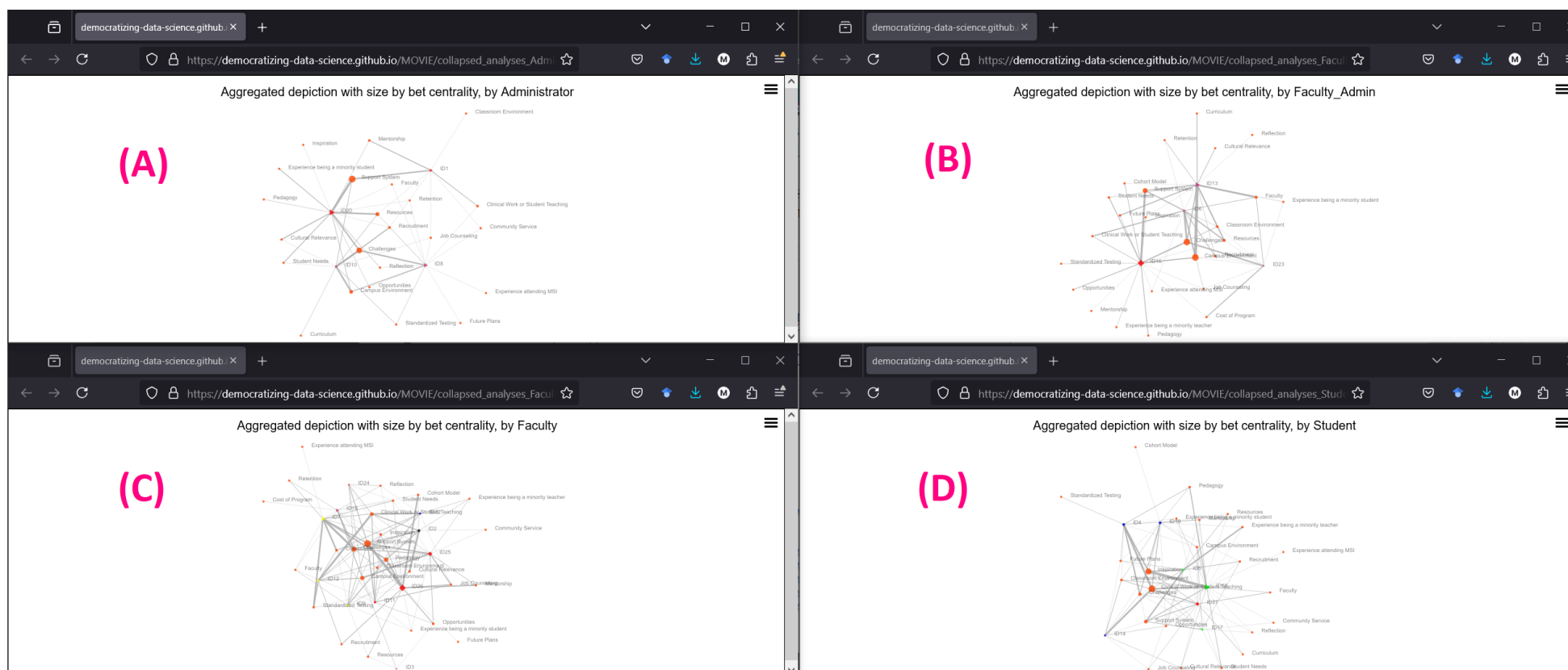


Figure 6. Disaggregated Sociograms.

HMTL Figure 6(a): <https://cutt.ly/GricQWKw>

HMTL Figure 6(b): <https://cutt.ly/DriOLXrJ>

HMTL Figure 6(c): <https://cutt.ly/9ricUqEW>

HMTL Figure 6(d): <https://cutt.ly/zricI5pb>

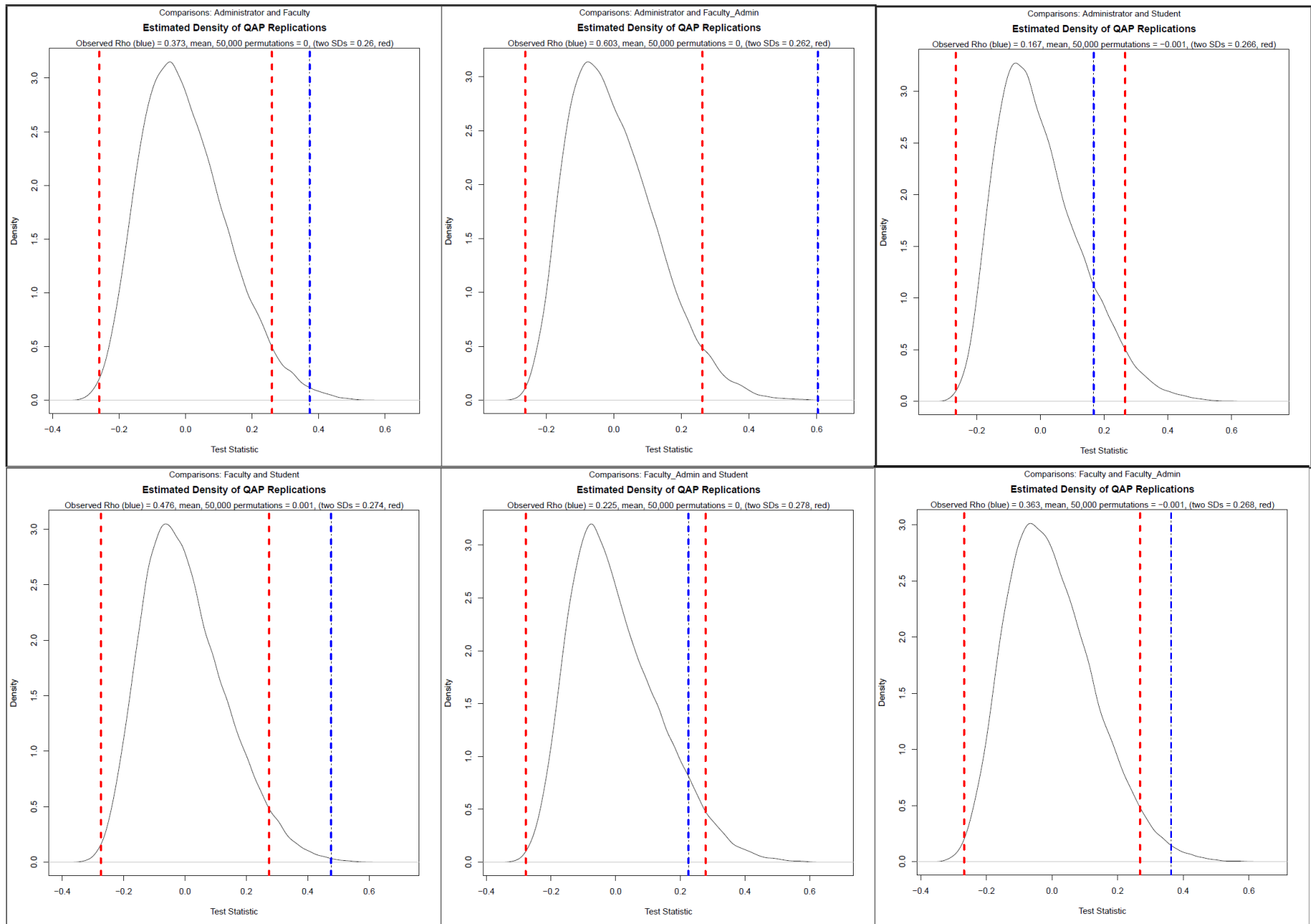


Figure 7. QAP Outputs

	A	B	C	D	E	F	G	H	I	J
1	names	communit	role	color	shape					
2	ID1	2	Administrator	color-based_on_community_assigned	man	Cross-Tabulation				
3	ID8	1	Administrator	color-based_on_community_assigned	man	Communities	Administrator	Faculty	Faculty_Admin	Student
4	ID10	1	Administrator	color-based_on_community_assigned	man	1	2	2	3	0
5	ID20	2	Administrator	color-based_on_community_assigned	man	2	2	4	1	1
6	ID2	3	Faculty	color-based_on_community_assigned	woman	3	0	1	0	0
7	ID3	1	Faculty	color-based_on_community_assigned	man	4	0	0	0	1
8	ID7	5	Faculty	color-based_on_community_assigned	man	5	0	3	0	0
9	ID9	5	Faculty	color-based_on_community_assigned	woman	6	0	1	0	5
10	ID11	2	Faculty	color-based_on_community_assigned	woman					
11	ID26	2	Faculty	color-based_on_community_assigned	woman					
12	ID12	5	Faculty	color-based_on_community_assigned	woman					
13	ID15	1	Faculty	color-based_on_community_assigned	man					
14	ID22	6	Faculty	color-based_on_community_assigned	woman					
15	ID24	2	Faculty	color-based_on_community_assigned	man					
16	ID25	2	Faculty	color-based_on_community_assigned	woman					
17	ID6	1	Faculty_Admin	color-based_on_community_assigned	man					
18	ID13	1	Faculty_Admin	color-based_on_community_assigned	man					
19	ID16	2	Faculty_Admin	color-based_on_community_assigned	woman					
20	ID23	1	Faculty_Admin	color-based_on_community_assigned	man					
21	ID4	6	Student	color-based_on_community_assigned	woman					
22	ID5	4	Student	color-based_on_community_assigned	man					
23	ID14	6	Student	color-based_on_community_assigned	woman					
24	ID17	2	Student	color-based_on_community_assigned	woman					
25	ID18	6	Student	color-based_on_community_assigned	man					
26	ID19	6	Student	color-based_on_community_assigned	woman					
27	ID21	6	Student	color-based_on_community_assigned	woman					
28										

Figure 8. Classified Database with Initial Cross-tabulations

NAQD [Network Analysis of Qualitative Data]
File Edit View Window Help

NAQD [Network Analysis of Qualitative Data]
I. Upload Data
II. Optional adjustments
III. Execute NAQD
IV. About NAQD

A. Load your data or select an example

Your data will be displayed after loading is completed

C. Required: Select column names from your csv file

Actor column (i.e., code provider or document ID)

Code column (i.e., code provided or text passage or excerpt)

D. Optional: If no role, color, or shape are needed don't modify

Column denoting *role* based on actor (or sender) column

Color based on community assignment (recommended) or column selected

Column for *shape* based on non-personal attributes column (recommended)

No. of groups for role variable (default 25):

Color transparency (1 is solid):

If ready, go to step III above to execute

NAQD handles numeric and alphanumeric inputs

B. Upload your data here or load example(s) as indicated in step A

Click to browse for your CSV file or simply drag it here with your mouse

MINIMAL information required:

Your data should have **at least** two columns **actors' or documents' IDs** and **codes** provided by or retrieved from them.

See data format example under section A.

OPTIONAL information based on actors' attributes or documents' characteristics. These pieces are used to ***contextualize*** the relationships and the content of codes

To use RQDA researchers need to upload a database with **'ID'** and **'Code'** as depicted in the example provided.

Section D may be populated with **'role'** that indicates participants' function (e.g., student, administrator, faculty) and, **when selected** will automatically conduct **QAPs** and produce disaggregated interactive network visualizations by **role category**.

Option **'color'** by default is assigned based on community detected using spin-glass modeling. However, researchers may select other options, such as gender, for example.

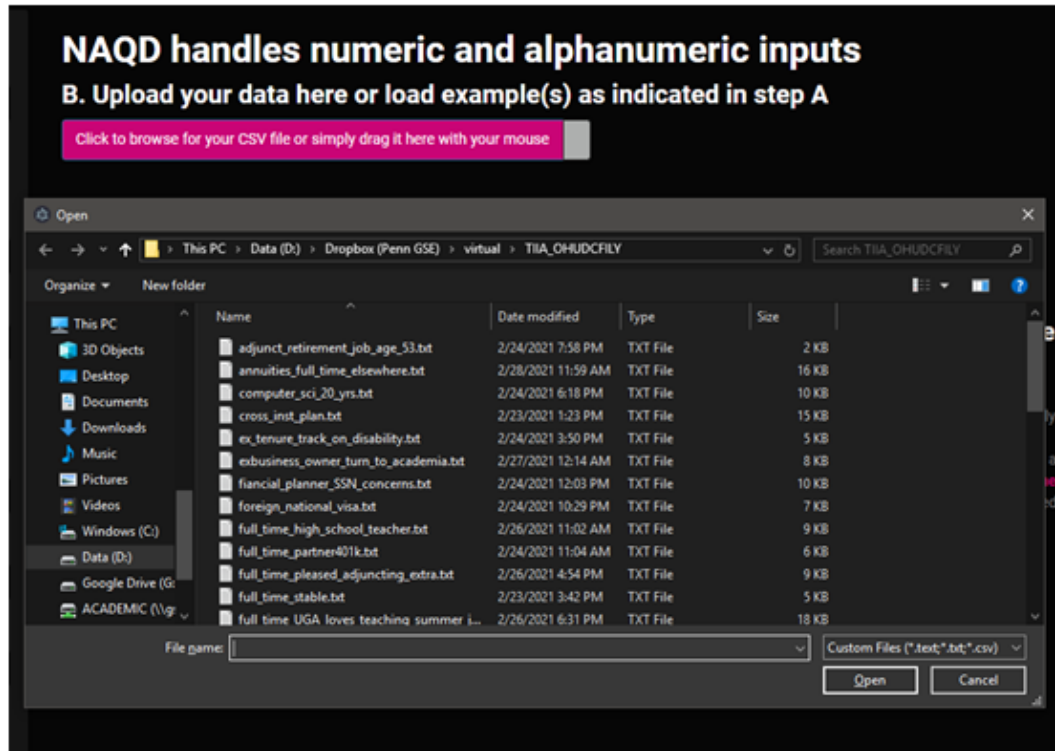
Option **'shape'** may be used to add other non-personal attributes of authors or documents' characteristics. **Retrieved by clicking on actors or documents** in the HTML visualizations.

Option **'No. of groups for role variable'** is the upper limit for the optimal number of communities. By default is selected 25 but can be increased or reduced based on network complexity.

Option **'Color transparency'** ranges from solid (1) to transparent (0) in the network visualizations.

Figure 9. NAQD User Interface

(A) Select data from your computer



(B) Load the example provided with NAQD

A. Load your data or select an example

Interview or multiple focus groups example

C. Required: Select column names from your csv file

Actor column (i.e., code provider or document ID)

Code column (i.e., code provided or text passage or excerpt)

D. Optional: If no role, color, or shape are needed don't modify

Column denoting "role" based on actor (or vendor) column

Column for "color" based on non-personal attributes column (recommended)

No. of groups for role variable (default 25)

Color transparency (1 to solid)

If ready, go to step III above to execute

NAQD handles numeric and alphanumeric inputs

B. Upload your data here or load example(s) as indicated in step A

Click to load interview data example shown below

MINIMAL information required:

Column: `text` and `code` and `code` (or `document`) (or `code` and `code`)

OPTIONAL information based on actors' attributes or documents' characteristics. These pieces are used to "contextualize" the relationships and the content of codes

In this example, `text` is the actor or document that was qualitatively coded.

`text` is the document text.

`code` is the content of each code.

`role` captures actor's role in the study (i.e., student, administrator, faculty, faculty-administrator).

`gender` is a personal attribute as reported by participants.

Option: No. of groups for role variable: In section II is the upper limit for the optimal number of communities. By default is selected 25 but can be decreased or increased based on network complexity.

Option: Color transparency: ranges from solid (1) to transparent (25) in the network visualizations.

Show 25 entries

ID	Code	Text	Role	Gender
1	001	Classroom Environment	Administrator	man
2	001	Critical Work or Student Teaching	Administrator	man

Figure 10. Data Handled by NAQD and Loading Process